Multi-Source Candidate Data Transformer

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1 — PIPELINE

Sources (CSV, GitHub) → **Parsers** (per-source, isolated) → **Normalizer** (phones, dates, skills, country) → **Identity Resolver** (explicit map → email fallback) → **Merge Engine** (conflict resolution + confidence) → **CanonicalProfile** (internal only) → **Projector** (runtime config) → **Validator** (schema check) → **Output JSON**

Each stage is a separate module. No stage imports from a non-adjacent stage. Adding a new source = one new Parser class, zero changes to downstream code. The CanonicalProfile never leaves the merge layer directly — all output is projected.

2 — CANONICAL SCHEMA

Field	Type	Notes
candidate_id	string	Stable, deterministic
full_name	string	
emails	string[]	Deduplicated
phones	string[]	E.164 format
location	object	{city, region, country: ISO-3166-a2}
links	object	{linkedin, github, portfolio, other[]}
headline	str | null	
years_experience	num | null	
skills	object[]	{name, confidence, sources[]}
experience	object[]	{company, title, start, end, summary}
education	object[]	{institution, degree, field, end_year}
provenance	object[]	{field, source, method, raw_value}
overall_confidence	float	0.0 – 1.0, weighted mean

3 — NORMALIZATION

- **Phones:** *phonenumbers* lib → E.164. Unparseable → null + raw preserved in provenance.
- **Dates:** ISO parse, then common formats (MM/YYYY, Month YYYY). Always YYYY-MM or null.
- **Skills:** Canonical map (js→JavaScript, golang→Go). Unknown skills kept as-is at lower confidence — never dropped.
- **Country:** *pycountry* → name, alpha-2, alpha-3. Unresolvable → null.

4 — IDENTITY RESOLUTION

Primary: Explicit mapping file (--id-map) pairing CSV row ID to GitHub username. Deterministic, auditable, zero ambiguity.

Fallback: Normalized email match when no mapping is provided.

Deliberate choice: Name-based fuzzy matching is intentionally excluded. The assignment warns that wrong-but-confident values are worse than honest nulls. A silent mismerge is a data integrity failure.

5 — MERGE & CONFLICT RESOLUTION

Each extracted value carries a **ValueEvidence** tuple: (value, source_id, method, confidence). Resolution priority:

- Structured source (CSV) beats unstructured (GitHub) for identity fields.
- Higher confidence wins when sources are of the same type.
- Most recent wins for time-sensitive fields (title, company).
- List fields (skills, emails): union + deduplicate. Every item retains its source.

All losing values are still written to provenance. Nothing is silently discarded.

6 — CONFIDENCE MODEL

Signal	Score
Structured, validated	0.90
Structured, unvalidated	0.75
Unstructured, clean	0.70
Unstructured, inferred	0.50
Single source only	-0.10
Corroborated by 2nd source	+0.10

overall_confidence = weighted mean of per-field scores. Name and email weighted higher than portfolio link. Fixed weights — explainability over flexibility.

7 — CONFIGURABLE PROJECTION

The projector is a **pure function**: (CanonicalProfile, ProjectionConfig) → dict. It has no knowledge of parsers or sources. Config supports:

- Field selection — emit only specified fields.
- Rename/remap — `from: "emails[0]"` → `path: "primary_email"`.
- Normalization hooks — per-field directives (E164, canonical).
- on_missing behavior — null | omit | error.
- Toggle provenance/confidence on or off.

Output is validated against the projected schema via Pydantic before returning. Errors are surfaced, never swallowed.

8 — EDGE CASES

Edge Case	Handling
GitHub API unreachable	Log warning, continue with CSV only
CSV row: no email, no ID map	Standalone record, flagged unresolved
Conflicting names	Structured wins; both in provenance
Unrecognized phone format	null; raw value in provenance
Skill in multiple sources	Union; highest confidence; all sources listed

9 — DELIBERATE DESCOPES

- PDF/resume parsing — brittle, not what this assignment evaluates.
- LinkedIn — no clean public API; ToS risk.
- Async fetching — valuable at scale; noted as future improvement.
- Fuzzy name matching — risk of silent mismerge outweighs benefit.